HH-M-391E
May 12, 1976
SUPERSEDING
Fed. Spec. HH-M-391D
February 14, 1969

FEDERAL SPECIFICATION

MITTENS, CLOTH, ASBESTOS (LEATHER REINFORCED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for ambidextrous asbestos mittens.

1.2 Classification. The mittens shall be of one type, and shall be procured in the following sizes as specified (see 6.2):

12 inch (30.5 cm) length
21 inch (53.3 cm) length

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276 - Thread, Cotton.
KK-L-162 - Leather, Cattlehide, Chrome Tanned for Gloves Garments, and Equipage.
SS-C-466 - Cloth, Thread, and Tape; Asbestos.
DDD-L-20 - Label; for Clothing, Equipage, and Tentage (General Use).

Federal Standards:

Fed. Std. No. 123 - Marking For Shipment (Civil Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.
Fed. Std. No. 751 - Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. FSC 8415

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-17757 - Boxes, Shipping, Fiberboard (Modular Sizes).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Postal Service Manual.

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U. S. Department of Agriculture:

Methods of Tests for Grades of Wool.

(Application for copies should be addressed to the U.S. Department of Agriculture, Agriculture Marketing Service, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

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(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to suppliers (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2).

3.3 Materials.

3.3.1 Cloth, asbestos. The asbestos cloth used for the outer shell of mittens (see 3.5.1), shall conform to form 1, grade U. G., style 1, of SS-C-466. This does not preclude the use of an additive finish as may be customary to reduce lint and dust during manufacturing operations. However, the additive shall have a minimum effect on the flexibility of the fabric and shall not decrease the asbestos fiber content below the minimum requirements.

3.3.2 Leather. The leather used for the reinforcement patches on outer shell of mittens (see 3.5.1), shall be flesh split cattlehide conforming to type IV of KK-L-162. The leather shall be natural light gray in color, and shall have a thickness of not less than 2-1/2 ounces (70 g) nor more than 3-1/2 ounces (98 g).

3.3.3 Hand liner. The inside of mittens shall have a knitted wool-cotton fabric liner, constructed with one or two ends of wool spun yarn tucked as one on the face and with one or two ends of cotton spun yarn knitted as one on the back, unbleached or dyed a suitable light gray or tan (see 3.5.1). The wool shall be made from fleece or pulled sheep's wool, or an combination thereof, not lower in grade than 50's U.S. Standard (see 4.3.2). The use of reprocessed, reused, or waste wool is prohibited. The cotton shall be of a suitable staple length and grade, and shall not be more than the 30 percent of the finished fabric weight. The knitted fabric shall be brushed or napped on the face side to obtain a finished uniform fleece effect, and shall conform to the physical requirements specified in Table I when tested as specified in 4.3.2.

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Table I - Physical Requirements

Characteristic	Requirement	
	Minimum	Maximum
Weight, ounces (g) per square yard (0.8 m^2)	9.0 (252 g)	11.0 (308 g)
Wales per inch (25.4 mm)	19	22
Courses per inch (25.4 mm)	17	23

3.3.4 Thread. The thread for seaming and stitching the mittens shall be unbleached cotton, conforming to type IA1, ticket no. 16, 4-ply, of V-T-276.

3.3.5 Label. Each mitten shall have a combination identification and size label conforming to type I, class 4, of DDD-L-20, except that the label shall bear the following inscription:

MITTENS, CLOTH, ASBESTOS, PLAINCUFF
LINER: 70% wool/30% COTTON (EXAMPLE)
CONTRACT NO: DSA-000-000-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
SIZE: 12 INCH (30.5 cm) LENGTH (EXAMPLE)
NAME OF SUPPLIER:

CAUTION
Contains Asbestos Fibers
Avoid Creating Dust
Breathing Asbestos Fibers May Create Serious
Bodily Harm

All capital letters specified for the cautionary statement shall be not less than 1/4 inch (6.35 mm) in height, and the lower case lettering not less than 1/8 inch (3.17 mm) in height. The printing on the label shall show fastness to accelerated laundering.

3.4 Design. The mitten shall be of an ambidextrous design, consisting of an outer asbestos shell with three leather reinforcement patches and a napped wool-cotton fleece liner (see Figures 1 and 2).

3.4.1 Figures. Figures 1 and 2 are furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figures, the written specification shall govern.

3.5 Patterns. Standard patterns to be used to cut the working patterns will be furnished by the Government (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way.

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3.5.1 Pattern parts. The component parts of the mittens shall be cut from materials as specified, and in accordance with the following pattern parts required for the manufacturing process:

Material	Nomenclature of Pattern Parts	Cut Parts per Pair
Cloth, asbestos (outer shell)	Hand	2
	Thumb	2
Cloth, wool-cotton fleece (hand liner)	Hand	2
	Thumb	2
Leather (reinforcement patches)	Finger	2
	Palm	2
	Thumb	2

3.6 Construction.

3.6.1 Stitches, seams, and stitchings. Stitch, seam, and stitching types specified in Table II, shall conform to FED-STD-751.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table II.

3.6.3 Type 301 stitching. Ends of 301 stitching shall be backstitched or backtacked not less than 3/8 inch (9.52 mm), except where ends are caught in other seams and stitching. Thread tension shall be maintained so there will be no loose stitching resulting in a loose bobbin or top thread or no excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn. The needle used shall be of the smallest size possible which will accomodate the specified size of thread.

3.6.3.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 3/8 inch (9.52 mm) back of the end of stitching.

b. Thread breaks, or two or more consecutive skipped stitches or run-offs noted during inspection of the item (in-process or end item), shall be repaired. The stitching shall start a minimum of 3/8 inch (9.52 mm) back of the defective area, continue over the defective area and continue a minimum of 3/8 inch (9.52 mm) beyond the defective area onto the existing stitching. Loose or tight stitching shall be repaired by removing the de-

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fective stitching without damaging the material and restitching in the required manner. 1/

1/ When making the above repairs, the ends of stitching are not required to be backstitched.

3.7 Manufacturing operations. The mittens shall be manufactured in accordance with operation requirements specified in Table II. The supplier is not required to follow the exact sequence of operations shown, provided that the finished mittens are identical to those produced by following the sequence as shown in Table II.

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Table II - Construction of the Mittens

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting a. Cut the component parts of the mittens in strict accordance with the patterns furnished, which show size, shape, directional lines, notches, and instructions for the proper assembly of all parts. b. All the fabric component parts shall be cut from one piece of material, with the length of the part running in the warp or wale direction as applicable.					
2.	Replacement of defective components During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in Section 4, shall be removed from production and replaced with non-defective and properly matched components.					
3.	Marking Mark, ticket, or bundle all component cut parts to insure a proper size assembly.					
4.	Attach leather reinforcement patches to outer shell a. Lap the leather thumb reinforcement patch over the thumb section of outer shell, and stitch the reinforcement patch to the thumb along the sides and around its curved portion 1/8 to 3/16 inch (3.17 to 4.76 mm) from the edge.	301	LSbj-1	9-11	16/4	16/4
	b. Lap the leather palm reinforcement patch over the hand section of outer shell, and stitch the reinforcement patch to the hand around all four sides 1/8 to 3/16 inch (3.17 to 4.76 mm) from the edge.	301	LSbj-1	9-11	16/4	16/4

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Table II - Construction of the Mittens (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Attach leather reinforcement patches to outer shell (cont'd)					
	c. Lap the leather finger reinforcement patch over the finger portion of outer shell, and stitch the straight edge of the reinforcement patch to the finger portion 1/8 to 3/16 inch (3.17 to 4.76 mm) from the edge.	301	LSbj-1	9-11	16/4	16/4
5.	Attach label to outer shell					
	Position and center the label specified in 3.3.5 between the leather palm reinforcement patch and the top edge of outer shell on each mitten, and stitch the label to the outer shell around all four sides not less than 1/16 inch (1.58 mm) from the edge. The label shall finish with the legend facing the finger portion of the outer shell.	301		9-11	16/4	16/4
6.	Assemble outer shell					
	a. Inseam stitch the thumb to the hand portion of mitten 3/8 to 1/2 inch (9.52 to 12.70 mm) from the edge, simultaneously catching the unsewn edges of the leather thumb and palm reinforcement patches.	301	SSa-1	9-11	16/4	16/4
	b. Fold the outer shell along the center to its finished position, and close the thumb by inseam stitching continuously around its outer periphery to the top open edge of outer shell 3/8 to 1/2 inch (9.52 to 12.70 mm) from the edge.	301	SSa-1	9-11	16/4	16/4

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Table II - Construction of the Mittens (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
6.	Assemble outer shell (cont'd) c. Close the outer shell by inseam stitching along the side and around its curved portion 3/8 to 1/2 inch (9.52 to 12.70 mm) from the beginning or ending at top open edge of shell.	301	SSa-1	9-11	16/4	16/4
7.	Assemble liner a. Inseam stitch the thumb to the hand portion of liner 3/8 to 1/2 inch (9.52 to 12.70 mm).	301	SSa-1	9-11	16/4	16/4
	b. Fold the liner along the center to its finished position, and close the thumb by inseam stitching continuously around its outer periphery to the top open edge of liner 3/8 to 1/2 inch (9.52 to 12.70 mm) from the edge.	301	SSa-1	9-11	16/4	16/4
	c. Close the liner by inseam stitching along the side and around the curved portion 3/8 to 1/2 inch (9.52 to 12.70 mm) from the edge beginning or ending at top open edge of liner. NOTE: The edges of all seams shall be trimmed approximately 1/8 inch (3.17 mm). A trimmer attachment may be used to accomplish this operation.	301	SSa-1	9-11	16/4	16/4
8.	Attach liner to outer shell Superimpose the liner to the outer shell, and tack the liner to the shell at finger tips for a distance of 1-1/2 to 2 inches (3.8 to 5.1 cm) and at thumb tip approximately 1 inch (2.5 cm), 1/8 to 3/16 inch (3.17 to 4.76 mm) from the edge.	301	SSa-1	9-11	16/4	16/4

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Table II - Construction of the Mittens (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
9.	Close mitten a. Turn the outer shell right side out by pulling it down over the liner, with the liner aligned smoothly within the shell.					
	b. Turn in raw edge of outer shell at top opening of mitten 3/8 to 1/2 inch (9.52 to 12.70 mm) and stitch around the circumference, 1/16 to 1/8 inch (1.58 to 3.17 mm) from the raw edge.	301	EFa-1	9-11	16/4	16/4
	c. Turn in raw edge of liner at top opening of mitten 3/8 to 1/2 inch (9.52 to 12.70 mm). Align the folded edge of the liner to the top opening of the outer shell, and stitch around the circumference 1/16 to 1/8 inch (1.58 to 3.17 mm) from the folded edge.	301	SSc-1	9-11	16/4	16/4
	d. When finished, the liner shall not protrude over the outer shell.					
10.	Cleaning a. Remove all spots and stains.					
	b. Trim all thread ends and remove all loose hanging thread.					
11.	Mating The finished mittens shall be mated in the applicable lengths, and tacked or stapled together in accordance with the best commercial practice which will permit the mittens to be separated without damaging the mitten. When staples are used, the metal shall be non-corrosive.					
a						

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3.8 Measurements. The finished mittens shall conform to the dimensions specified in Table III. All measurements and tolerances are expressed in inches. To convert to metric system, multiply inches (decimal equivalent) by 2.54 millimeters.

Table III - Finished measurements

Measurement	Size		Tolerance
	12 inch	21 inch	
Overall length 1/	12	21	± 1/4
Width of cuff 2/	6-1/4	8	+ 3/8 - 1/4
Width of fingers 3/	5-3/4	5-3/4	+ 3/8 - 1/4

1/ Measurement shall be taken in a perpendicular line from tip of fingers to top open edge of mitten.

2/ Measurement shall be taken along top open edge of mitten, folded edge to folded edge.

3/ Measurement shall be taken along the horizontal seam of leather reinforcement patch next to the fingers, folded edge to folded edge.

3.9 Workmanship. The finished mittens shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.3 Inspection. Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

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4.3.1 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are performed as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance.

4.3.2 Component and material inspection. In accordance with 4.1, components and material shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, the cotton and wool fleece material shall be tested for the characteristics specified in Table IV and in accordance with the referenced test methods of FED-STD-191. Testing for all the requirements listed in Table IV is applicable to the sample unit. The lot size shall be expressed in units of pounds (kg) and the sample unit shall be 1/2 yard (0.45 m), full width. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any requirements specified.

Table IV - Component testing

Characteristic	Requirement paragraph	Test method
Material identification of face yarn	3.3.3	1100 1/
Type of wool	3.3.3	1/
Wool grade	3.3.3	1/
Material identification of backing yarn	3.3.3	1200 1/
Type of knit construction	3.3.3	Visual 2/
Percentage of cotton in fabric	3.3.3	2100 3/
Wales per inch (25.4 mm)	3.3.3	5070
Courses per inch (25.4 mm)	3.3.3	5070
Weight	3.3.3	5041

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

3/ The wool content shall be obtained first, based on the dry weight of the specimen. The remainder shall be considered as the cotton percent.

4.3.3 Intermediate examination. Examination of the completely assembled knit liner shall be made prior to attachment to the outer shell. The inspection level shall be I and the acceptable quality levels shall be 2.5 major defects and 6.5 total (major and minor defects combined) defects per hundred units. The sample unit shall be one knit liner. The lot size shall be expressed in terms of one knit liner each.

4.3.3.1 Visual xamination

Defects	Classification Major	Minor
Material Defects and Damages		
a. Hole, cut, tear or rip, or needle chew(s).	X	
b. Broken yarn, dropped stitch or run.	X	
c. Loose yarn, pulled or snagged yarn.		X
d. Any visible mend (not applicable to a restitched seam).		X
e. Not napped on the face side of the material.	X	
f. Any visible un-napped area on the face side of the material.		X
Color		
Not as specified.		X
Cutting		
Any component part not cut in accordance with specified pattern, directional lines on patterns, or not in accordance with specification requirements.	X	
Components and Assembly		
a. Any component part or operation omitted.	X	
b. Any component not as specified.	X	
c. Any component part misplaced or not serving intended purpose.		X
d. Any component part not securely affixed, e.g., thumb piece loosely attached to hand portion of liner.		X
e. Any workmanship damage, e.g., scissor or knife cut.	X	

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Defects	Classification Major	Minor
Seams and Stitching		
a. Any open seam.	X	
NOTE: The seam shall be classified open when one or more stitches joining a seam are broken or when two or more continuous skipped or run-off stitches occur.		
b. Repairs of open seams not as specified.		X
c. Not specified seam or stitch type.	X	
d. Any line of stitching omitted.	X	
e. Any line of stitching misplaced or not beginning or ending where specified.	X	
f. Any part of liner caught in an unrelated operation.	X	
g. Loose tension resulting in a loose seam.	X	
h. Tight tension (stitches break or cutting of the material results when normal strain is applied in the direction of the seam).	X	
i. End of 301 stitching not backstitched when not caught in another row of stitching, or backstitched less than 3/8 inch (9.52 mm).		X
j. Stitches per inch (to be scored only when the condition exists on the major portion of the seam).		
1. One or two stitches per inch less than the minimum.		X
2. Three or more stitches per inch less than the minimum.	X	
3. More than the specified maximum number of stitches per inch, damaging the assembly.	X	
4. More than the specified maximum number of stitches per inch, but not damaging the assembly.		X
k. Gauge of stitching irregular or not as specified:		
1. Seriously affecting serviceability.	X	
2. Affecting serviceability, but not seriously.		X
Assembly Detail		
a. Thread ends not trimmed throughout the liner or loose thread not removed.		X
b. Any thumb or remaining finger areas distorted or twisted, affecting comfort of wearer.	X	
c. Not constructed with specified number of pieces for the liner.	X	

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4.3.4 Examination of the end item. The defects found during the examination of the end item shall be classified in accordance with 4.3.4.1 and 4.3.4.2. The applicable inspection levels and acceptable quality levels (AQL's), shall be as indicated in 4.3.4.3. The lot size shall be expressed in units of one mitten each. The sample unit shall be one mitten and selection shall be by pairs. Defects for mating shall be classified as a single defect.

4.3.4.1 Visual examination.

Defects	Classification Major	Minor
Material Defects and Damages (leather)		
a. Not type specified.	X	
b. Any hole (except pinhole or needle hole), cut tear or rip.	X	
c. Brittle or bony area; thin or weak area:		
- Seriously affecting serviceability.	X	
- Affecting serviceability, but not seriously.		X
d. Imperfection(s), e.g., grub or tick marks, scratch, scar, or abrasions:		
- Seriously affecting serviceability.	X	
- Affecting serviceability, but not seriously.		X
e. Three or more needle holes or pinholes.	X	
f. Loose, spongy, or flanky:		
- Seriously affecting serviceability.	X	
- Affecting serviceability, but not seriously.		X
g. Open vein or rupture.	X	
h. Color not as specified.		X
Material Defects and Damages (asbestos outer shell)		
a. Hole, cut, tear or rip, or needle chew(s).	X	
b. Not type specified.	X	
c. Any visible mend (not applicable to a restitched seam).		X
d. Any weakened area.	X	
Cutting		
Any component part not cut in accordance with specified pattern, directional lines on patterns, or not in accordance with specification requirements.	X	

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Defects	Classification Major	Minor
Components and Assembly		
a. Any component part or operation omitted.	X	
b. Any component not as specified.	X	
c. Any component part misplaced or not serving intended purpose.		X
d. Any component part not securely affixed, e.g., thumb piece loosely attached to hand portion of outer shell.		X
e. Any workmanship damage, e.g., scissor or knife cut.	X	
Seams and Stitching		
a. Any open seam. NOTE: The seam shall be classified open when one or more stitches joining a seam are broken or when two or more continuous skipped or run-off stitches occur.	X	
b. Repairs of open seams not as specified.		X
c. Not specified seam or stitch type.	X	
d. Any line of stitching omitted.	X	
e. Any line of stitching misplaced or not beginning or ending where specified.	X	
f. Any part of mitten caught in an unrelated operation.	X	
g. Loose tension resulting in a loose seam.	X	
h. Tight tension (stitches break or cutting of the cloth or leather results when normal strain is applied in the direction of the seam).	X	
i. End of 301 stitching not backstitched when not caught in another row of stitching, or backstitched less than 3/8 inch (9.52 mm).		X
j. Stitches per inch (to be scored only when the condition exists on the major portion of the seam).		
1. One or two stitches per inch less than the minimum.		X
2. Three or more stitches per inch less than the minimum.	X	
3. More than the specified maximum number of stitches per inch, damaging the assembly.	X	
4. More than the specified maximum number of stitches per inch, but not damaging the assembly.		X

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Defects	Classification Major	Minor
Seams and Stitching (cont'd)		
k. Gauge of stitching irregular or not as specified:		
1. Seriously affecting serviceability.	X	
2. Affecting serviceability, but not seriously.		X
Assembly Detail		
a. Thread ends not trimmed or loose thread not removed.		X
b. Any thumb or remaining finger areas distorted or twisted, affecting comfort of wearer.	X	
c. Not constructed with specified number of leather reinforcement patches.	X	
Pairing		
a. Mismated, i.e., two mittens of different lengths or varied in color.		X
b. Pair not tacked or stapled together.		X
Label		
a. Omitted, incorrect, or illegible.	X	
b. Misplaced, not attached as specified, or stitching through the printing.		X

4.3.4.2 Dimensional examination. The mittens shall be examined for dimensional defects. Any dimension that is not within the established tolerance, shall be classified as a defect (see 3.8).

4.3.4.3 Inspection levels and acceptable quality levels (AQL's). The inspection levels, expressed in defects per 100 units for visual and dimensional examination, shall be as follows:

	Inspection level	AQL's Major	Total
For defects applicable to 4.3.4.1	II	2.5	6.5
For defects applicable to 4.3.4.2	S-3	-	4.0

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4.3.5 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with the Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling; bulging or distortion of container.
Content	Pairs of mittens per bundle is more or less than specified. 1/ Number of bundles per shipping container is more or less than specified.

1/ For this defect, two bundles from each shipping container in the sample shall be examined.

5. PREPARATION FOR DELIVERY

5.1 Preservation - packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 12 inch (30.5 cm) mittens. Nine (9) pairs of mittens shall be evenly stacked in a bundle with each pair reversed end to end, and securely tied at each end with cotton tape or twine.

5.1.1.2 21 inch (53.3 cm) mittens. Each pair of mittens shall be folded in half to an approximate length of 11-1/2 inches (29.2 cm). Six (6) pairs of mittens shall be evenly stacked in a bundle with each pair reversed end to end, and securely tied at each end with cotton tape or twine.

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5.1.2 Level B (Civil agencies). The 12 inch (30.5 cm) and the 21 inch (53.3 cm) mittens shall be packaged in accordance with 5.1.1.1 and 5.1.1.2 respectively (see 6.2).

5.1.3 Level C (Commercial packaging). Mittens shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Thirty-six (36) pairs of the 12 inch (30.5 cm) mittens (4 bundles of 9 pairs each); or twenty-four (24) pairs of the 21 inch (53.3 cm) mittens (4 bundles of 6 pair each); packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to class weather-resistant, grade V2s, size 3A, of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c, of MIL-B-17757. Level A packages shall be packed flat, two in length and two in width, with the length of the packages parallel to the side panels of the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of kraft paper of any suitable commercial grade. Toward the end of the contract, or when there are less than the required amount per shipping container of the same size, mixed sizes may be packed within the same container.

5.2.2 Level B. Thirty-six (36) pairs of the 12 inch (30.5 cm) mittens (4 bundles of 9 pairs each); or twenty-four (24) pairs of the 21 inch (53.3 cm) mittens (4 bundles of 6 pairs each); packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to class domestic, grade 275, size 3A, of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 275 of MIL-B-17757. Level A packages shall be packed flat, two in length and two in width, with the length of the packages parallel to the side panels of the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of kraft paper of any suitable commercial grade. Toward the end of the contract, or when there are less than the required amount per shipping container of the same size, mixed sizes may be packed within the same container.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the container specification.

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5.2.3 Level C (Commercial packing). Mittens, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such deliveries. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2.1 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and sides directly under the printing or stenciling, a white paper label 5 by 4 inches (12.7 by 10.2 cm) with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

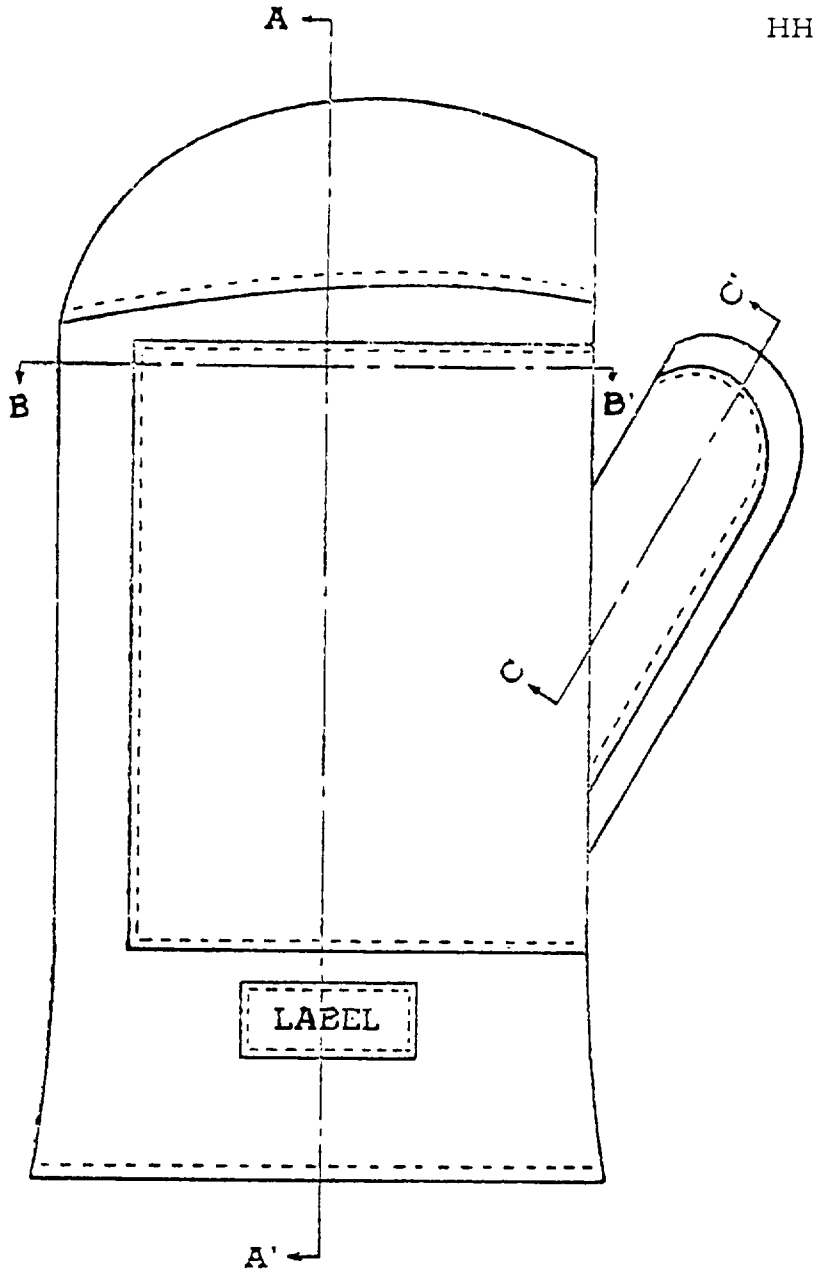
6. NOTES

6.1 Intended use. The mittens covered by this specification are intended for general use by Civil and Military government personnel in foundries and metal heat-treatment plants in order to protect hands and forearms from severe heat.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Whether first article sample is required (see 3.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. When Level B packaging is required for civil agencies (see 5.1.2).
f. When weather-resistant grade fiberboard shipping containers are required for Level B packing (see 5.2.2.1).

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NOTE: OTHER SIDE OF MITTEN SIMILAR TO SIDE SHOWN (EXCEPT FOR LABEL).

SEE FIGURE 2 FOR SECTIONS SHOWN.

Figure 1 — MITTENS, CLOTH, ASBESTOS (LEATHER REINFORCED)

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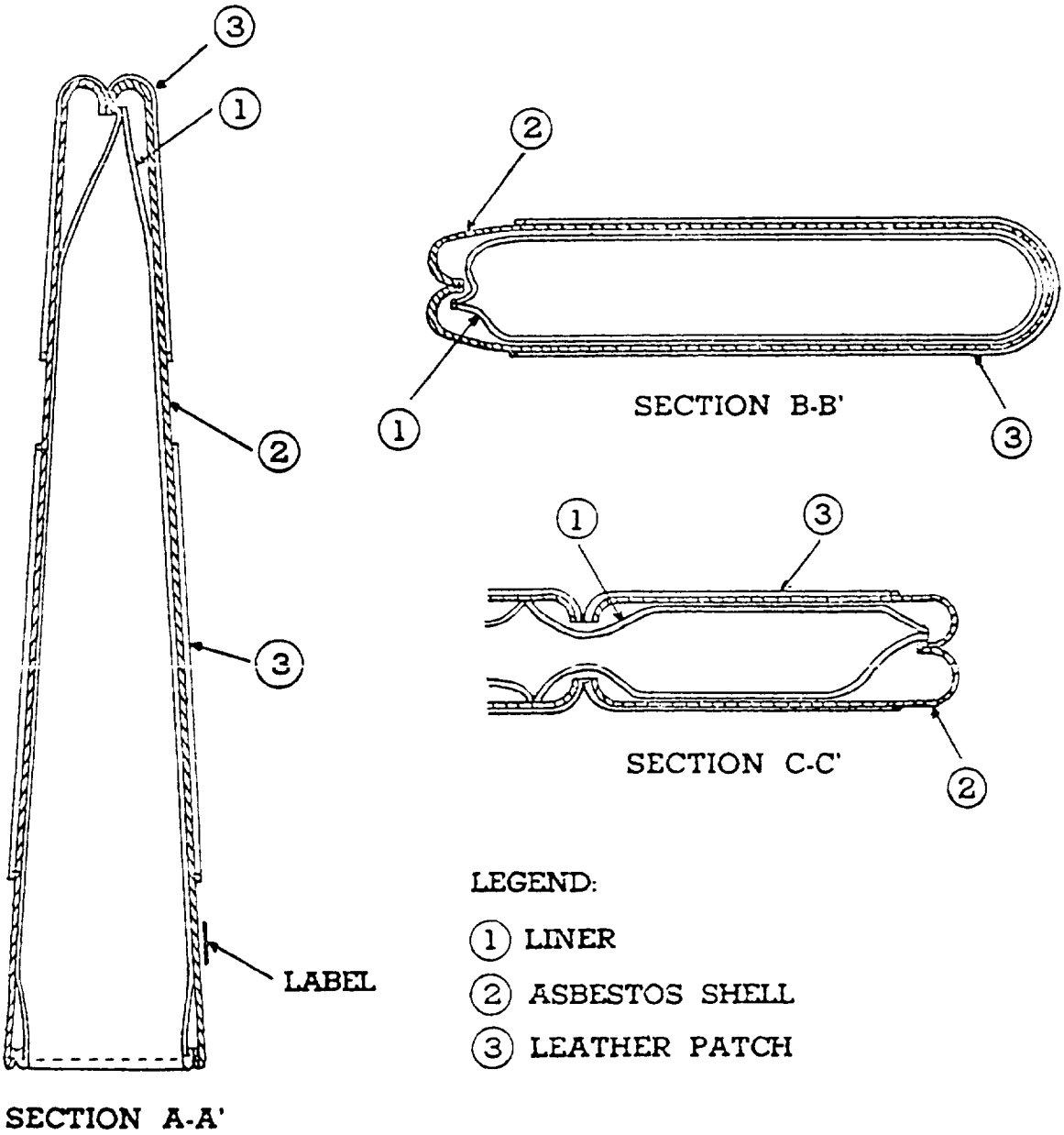


Figure 2 — SECTIONS

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6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

Military Custodian:
Navy - SA

User Activity:
Navy - YD

Preparing Activity
Navy - SA

Civil Agency Coordinating Activities:
GSA-FSS
TVA

Project No. 8415-0883

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 50 cents each.

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DOCUMENT IDENTIFIER AND TITLE

HH-M-391E - Mittens, Cloth, Asbestos (Leather Reinforced)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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	DATE

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